

MechMavrix CivicRoute - 1-Page Decision Log

Hackathon: HackMysuru 1.0 **Track:** Civic Governance & Clean Mysuru **Primary Subproblem:** Routing

Core decision: Focus deeply on **routing**, not a broad grievance portal. Citizens provide evidence and GPS; CivicRoute determines issue, urgency, authority, department and follow-through.

Key decisions and trade-offs

#	Decision	Choice	Why	Trade-off
1	Minimal citizen input	Photo + GPS + optional description only.	Citizens may not know jurisdiction, department, severity or priority.	Backend must perform more verification and handle uncertainty.
2	Server-side decision context	Backend derives address, nearby risk, routing and priority.	Reduces client-side manipulation and inconsistent decisions.	Depends on external map services while online.
3	Transparent 50/30/20 priority	Location 50, problem 30, severity 20.	Same issue can be more dangerous near hospitals, schools, traffic signals, bus stations or markets.	Weights are prototype policy values, not municipal policy.
4	AI assists; humans handle uncertainty	CLIP estimates issue and severity; low-confidence -> REVIEW REQUIRED.	Avoids forcing uncertain AI outputs.	Some complaints require manual review.
5	Versioned jurisdiction rules	Rules carry effective dates and provenance metadata.	Boundaries and administrative ownership can change.	MVP uses locality rules, not authoritative GIS polygons.
6	Conservative duplicate handling	Exact-image nearby reports may reuse an existing case; nearby matches are review candidates.	Reduces duplicate inflation without labeling citizens as spam.	Edited/cropped images may bypass exact hashing.
7	Offline-first preservation	Service Worker + IndexedDB queue + automatic sync.	Weak connectivity should not cause report loss.	Final AI/routing waits until backend reconnection.
8	Persistent follow-through	CR-* ID, citizen tracking, staff workflow and complaint history.	Routing without visibility does not solve follow-through.	Staff dashboard is demo-only; no production auth/RBAC yet.

We deliberately did not claim	Next production steps
- Real-time crowd density - Official/legal GIS boundary proof - Production AI accuracy percentage - Production-grade staff authentication - Existing municipal-system integration	Official GIS/PostGIS routing; authenticated staff roles; secure object storage; rate limiting; calibrated civic AI evaluation; SLA/escalation rules; resolution evidence verification.